Controlled enzyme statistical analysis

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Load packages

```
library(tidyverse)
library(lme4)
library(lmerTest)
library(emmeans)

dir.create("Controlled enzyme results", showWarnings = FALSE)
```

Import data

```
df <- read.csv("all_enzyme_data_controlled.csv")
```

Prepare variables

```
df$Species <- as.factor(df$Species)

df$Treatment <- as.factor(df$Treatment)

df$`Plant.part` <- as.factor(df$`Plant.part`)

df$Enzyme.function <- as.factor(df$Enzyme.function)

df$Enzyme <- as.factor(df$Enzyme)
```

Model

```
model <- lmer(
  `Log2.Fold.Change` ~
    Enzyme.function *
    Treatment *
    Species *
    `Plant.part` +
    (1 | Enzyme.function:Enzyme),
  data=df
)
```

ANOVA

```
anova_results <- anova(model)

anova_results
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                                           Sum Sq Mean Sq NumDF DenDF F value
## Enzyme.function                            4.475   1.119     4    13  0.3861
## Treatment                                  1.959   1.959     1    91  0.6761
## Species                                    3.462   3.462     1    91  1.1951
## Plant.part                                82.313  82.313     1    91 28.4118
## Enzyme.function:Treatment                  2.962   0.741     4    91  0.2556
## Enzyme.function:Species                   11.410   2.852     4    91  0.9846
## Treatment:Species                          2.304   2.304     1    91  0.7953
## Enzyme.function:Plant.part                13.708   3.427     4    91  1.1829
## Treatment:Plant.part                      10.431  10.431     1    91  3.6003
## Species:Plant.part                        12.387  12.387     1    91  4.2758
## Enzyme.function:Treatment:Species          1.212   0.303     4    91  0.1046
## Enzyme.function:Treatment:Plant.part       4.157   1.039     4    91  0.3587
## Enzyme.function:Species:Plant.part        15.845   3.961     4    91  1.3673
## Treatment:Species:Plant.part              10.078  10.078     1    91  3.4787
## Enzyme.function:Treatment:Species:Plant.part  6.228   1.557     4    91  0.5375
##                                              Pr(>F)
## Enzyme.function                             0.81478
## Treatment                                   0.41309
## Species                                     0.27720
## Plant.part                                7.077e-07 ***
## Enzyme.function:Treatment                   0.90554
## Enzyme.function:Species                     0.42007
## Treatment:Species                           0.37486
## Enzyme.function:Plant.part                  0.32368
## Treatment:Plant.part                        0.06094 .
## Species:Plant.part                          0.04150 *
## Enzyme.function:Treatment:Species           0.98065
## Enzyme.function:Treatment:Plant.part        0.83734
## Enzyme.function:Species:Plant.part          0.25150
## Treatment:Species:Plant.part                0.06539 .
## Enzyme.function:Treatment:Species:Plant.part   0.70856
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
write.csv(
    anova_results,
    "Controlled enzyme results/ANOVA_Table.csv"
)
```

Estimated marginal means

Treatments

```
emm_2 <- emmeans(model,  ~ Treatment | Enzyme.function * Species * Plant.part)

emm_pairs_2 <- pairs(emm_2, adjust = "BH")
emm_pairs_2
```

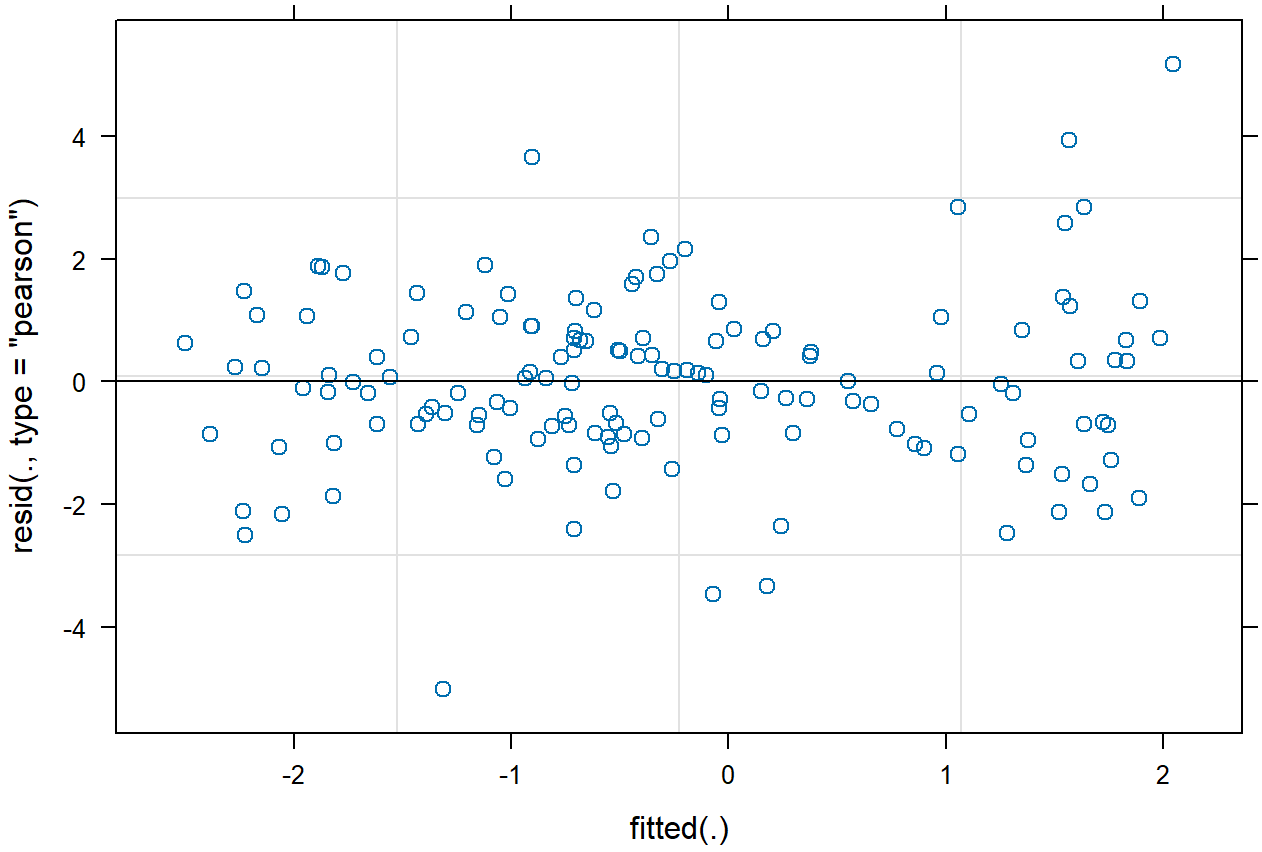
```
## Enzyme.function = Antioxidant, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -0.4790 0.983 91  -0.487  0.6271
##
## Enzyme.function = Carbon redistribution, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -0.5944 1.390 91  -0.428  0.6699
##
## Enzyme.function = Carbon storage, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -0.0570 1.390 91  -0.041  0.9674
##
## Enzyme.function = Glycolysis, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   0.0502 1.390 91   0.036  0.9713
##
## Enzyme.function = Sucrolytic activity, Species = D. flexuosa, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   0.9300 1.390 91   0.669  0.5051
##
## Enzyme.function = Antioxidant, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -0.2791 0.983 91  -0.284  0.7770
##
## Enzyme.function = Carbon redistribution, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   0.2850 1.390 91   0.205  0.8380
##
## Enzyme.function = Carbon storage, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -2.2879 1.390 91  -1.646  0.1032
##
## Enzyme.function = Glycolysis, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -1.3414 1.390 91  -0.965  0.3370
##
## Enzyme.function = Sucrolytic activity, Species = E. vaginatum, Plant.part = Root:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   0.6037 1.390 91   0.434  0.6651
##
## Enzyme.function = Antioxidant, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   0.7473 0.983 91   0.760  0.4490
##
## Enzyme.function = Carbon redistribution, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -0.1901 1.390 91  -0.137  0.8915
##
## Enzyme.function = Carbon storage, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -0.5445 1.390 91  -0.392  0.6962
##
```

```
## Enzyme.function = Glycolysis, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   0.1483 1.390 91   0.107  0.9153
##
## Enzyme.function = Sucrolytic activity, Species = D. flexuosa, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)  -0.2159 1.390 91  -0.155  0.8769
##
## Enzyme.function = Antioxidant, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   1.5150 0.983 91   1.542  0.1266
##
## Enzyme.function = Carbon redistribution, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   1.1103 1.390 91   0.799  0.4264
##
## Enzyme.function = Carbon storage, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   2.3168 1.390 91   1.667  0.0989
##
## Enzyme.function = Glycolysis, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   1.3930 1.390 91   1.002  0.3188
##
## Enzyme.function = Sucrolytic activity, Species = E. vaginatum, Plant.part = Shoot:
##  contrast                    estimate    SE df t.ratio p.value
##  R chilling - (R+S chilling)   1.7377 1.390 91   1.250  0.2144
##
## Degrees-of-freedom method: kenward-roger
```

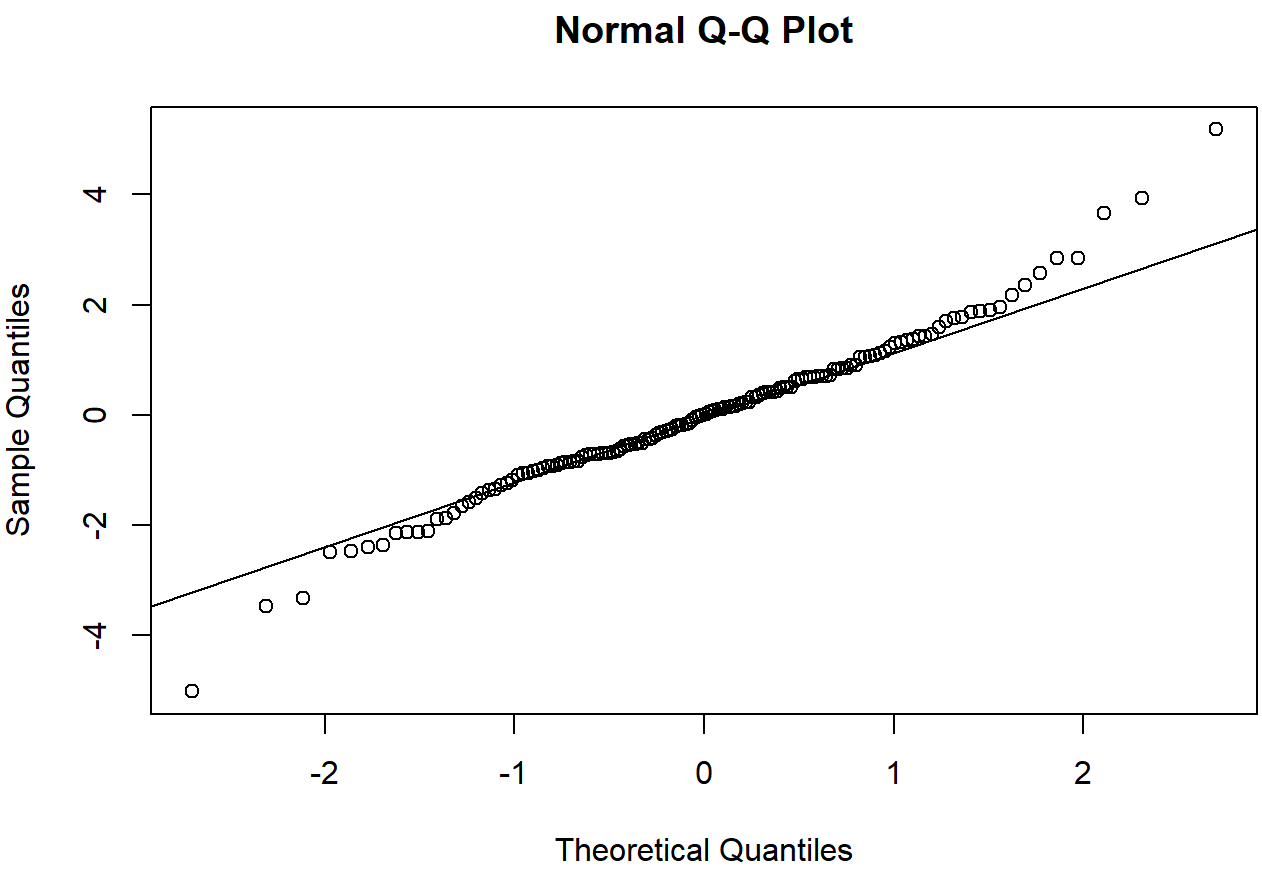
```
write.csv(emm_pairs_2,
          "Controlled enzyme results/Treatments_Pairwise_Comparisons.csv",
          row.names = FALSE)
```

Diagnostics

```
plot(model)
```




```
qqnorm(residuals(model))
qqline(residuals(model))
```


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
-4
-2
0
2
4
-2
-1
0
1
2